InternsElite — Data Science & Business Analytics Traineeship

Minor Project Report — Exploratory Data Analysis (EDA)

Spotify Track Dataset | Prepared from: minor_project.ipynb | September 2026

Project Title & Objective

Exploratory Data Analysis of a Spotify Track Dataset. The objective is to load, clean, and preprocess an 85,000-row music-track dataset and analyze it to uncover patterns and relationships among audio features (danceability, energy, tempo), popularity, and stream count.

Dataset Description

The dataset contains 85,000 tracks and 33 columns, including track/artist/album metadata, audio features (danceability, energy, loudness, tempo, instrumentalness), popularity score, stream_count, and several engineered fields (popularity_category, log_stream_count, upbeat_score, release_year/month/quarter, is_weekend_release). The data was found to have 0 missing values and 0 duplicate rows or track IDs. Column names were standardized (lower-case, underscores) and release_date was converted to a proper datetime type.

Tools and Libraries Used

Python (Jupyter Notebook) · Pandas · Matplotlib · Seaborn — for data loading, cleaning, quality checks, and visualization.

Screenshots of Key Visualizations

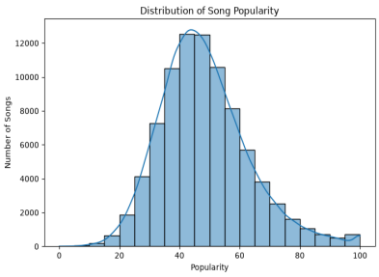


Fig 1: Popularity distribution

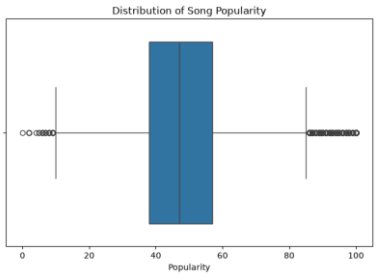


Fig 2: Popularity boxplot

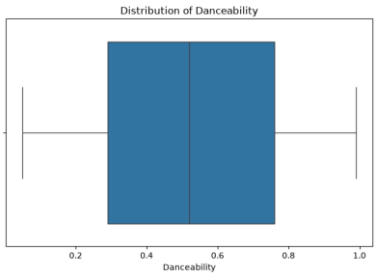


Fig 3: Danceability boxplot

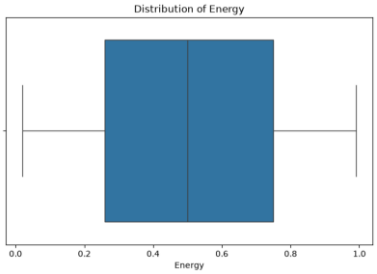


Fig 4: Energy boxplot

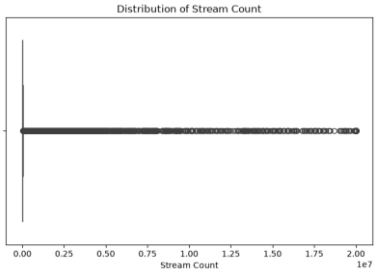


Fig 5: Stream count boxplot

Additional charts (correlation heatmap, genre-wise bar chart, scatter plots) are recommended as next steps — see Challenges section.

Summary of Insights

- Popularity is approximately normally distributed, peaking in the 40–50 range, with a small share of tracks as low/high-popularity outliers.
- Danceability and energy are both spread fairly evenly across their full 0–1 scale, with no significant outliers.
- Stream count is heavily right-skewed — most tracks have low streams while a long tail reaches up to ~20 million, which is why a log_stream_count feature was engineered.
- The dataset is fully clean (no nulls/duplicates), so effort focused on structural checks and distribution analysis rather than imputation.

Challenges Faced and Solutions

- Challenge: df.boxplot(columns='stream_count') raised a TypeError. Solution: corrected the keyword to column= (singular).
- Challenge: stream_count's extreme right-skew and outliers compressed the box plot. Solution: a log_stream_count column was engineered for better visual scaling.
- Challenge: the current notebook has 5 visualizations, below the 10-chart minimum in the guidelines. Solution/next step: add a correlation heatmap, genre-wise bar chart, popularity-vs-streams scatter plot, and time-trend line chart to meet the requirement.